Multi-level Domain Quiz Application

A project report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF COMPUTER APPLICATIONS

BY

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Mahatma Education Society's

PILLAI COLLEGE OF ARTS, COMMERCE & SCIENCE (Autonomous)

Re-accredited "A" Grade by NAAC (3rd Cycle)

CERTIFICATE

This is to certify that the project entitled "**Multi-level Domain Quiz Application**", is Bonafide work of **Ms. Mamta Tikuram Kurdia** bearing Seat No: 5841 submitted in partial fulfillment of the requirements for the award of degree of **BACHELOR OF COMPUTER APPLICATION** from University of Mumbai.

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Finally, I owe my deepest gratitude to my family for their unwavering encouragement and patience, and belief in me, which inspired me to persevere and complete this project successfully.

Yours faithfully,
Mamta Kurdia
(S.Y.BCA.A)

DECLARATION

I hereby declare that this project report entitled "**Multi-level Domain Quiz Application**" submitted by me to University of Mumbai is a record of my original work carried out under the guidance of Prof. Kshitiz Tiwari. This project has not been submitted to any other institution or university for the award of any degree, diploma, or certificate.

I further declare that all sources of information and assistance have been duly acknowledged in the report.

Date: Signature of Student

Place: New Panvel Name of Student: Mamta Tikuram Kurdia

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1. ABSTRACT

The **QuizMaster** application is a comprehensive digital learning and evaluation platform designed to streamline the process of knowledge assessment for both students and administrators. In the modern educational landscape, there is a growing need for interactive tools that provide immediate feedback and robust data management. This project addresses these needs by implementing a full-stack web application using the **MERN** (MySQL, Express.js, React, Node.js) architecture.

The application features a user-friendly frontend that allows students to engage with quizzes across various categories, featuring a real-time progress tracker, dynamic timers, and instant scoring. "Additionally, the system features an automated credentialing engine that generates a downloadable PDF certificate of completion for users who master the advanced difficulty tier, providing a formal record of their achievement." On the backend, a secure administrative dashboard was developed to provide oversight of student performance. Key administrative functionalities include the ability to monitor total users, track attempt history, calculate average scores, and perform critical data management tasks such as resetting student progress or deleting user accounts to maintain database integrity.

The system utilizes a relational database (MySQL) to ensure data consistency and employs modern web practices like theme toggling and responsive design to enhance the user experience. The results of this project demonstrate an efficient solution for academic evaluation, offering a scalable bridge between interactive learning and administrative management.

2. LIST OF SCREENS

1. **Landing Page (Home):** The entry point of the application. It features a professional hero section, an "About" overview of the platform, and clear Call-to-Action (CTA) buttons for users to "Get Started" or "Login." It introduces the concept of multi-level domain mastery.
2. **Login Page:** A secure authentication interface where existing users enter their credentials. It includes form validation and a toggle to redirect to the Signup page if the user doesn't have an account.
3. **Signup Page:** The registration module that collects user details (Username, Email, Password). Upon submission, the data is validated and stored in the MySQL users table, and the user is redirected to the dashboard.
4. **Admin Panel (Management Console):** A restricted-access dashboard for administrators. It features three main management modules:
    - **Domain Management:** To add or edit Main Categories.
    - **Topic Management:** To add Sub-categories to specific domains.
    - **Question Bank:** A CRUD interface to add multi-level questions (Easy to Advance).
    - **Analytics:** A visual summary of total users, active quizzes, and completion rates.
5. **User Home (Domain Dashboard):** The primary navigation hub for students. It fetches all "Main Domains" (e.g., Programming, Science) from the database and displays them as interactive cards with icons and descriptions.
6. **Sub-Category Page (Topic & Level Selection):** This screen displays specific topics within a chosen domain. Crucially, it shows the **Level Locking Logic**, where levels (Easy, Moderate, Hard, Advanced) are enabled or disabled based on the user's current progress fetched from the user_progress table.
7. **Quiz Page (Play Module):** the interactive assessment interface. It displays one question at a time with four options. It includes:
    - A progress bar or question counter.
    - An instant feedback system for correct/incorrect choices.
    - A final "Result Modal" with the score and a **"Download Certificate"** button for the Advanced level.
8. **User Progress (Achievements):** A personalized dashboard where users can view their journey. It displays stats like "Total Quizzes Taken," "Domains Mastered," and a history of high scores across different sub-categories.
9. **User Settings (Profile Management):** Allows users to update their profile information, such as changing their display name (which reflects on the certificate) or updating their password and profile avatar.

3. INTRODUCTION

The rapid evolution of information technology has fundamentally reshaped the educational landscape, shifting the focus from traditional pedagogical methods to interactive, digital-first learning environments. In this context, web-based assessment tools play a critical role in providing immediate evaluation and feedback to learners, allowing for a more personalized and paced learning experience.

This project, titled **DomainWise - Multi-Level Quiz Web App (QuizMaster)**, is a sophisticated full-stack application designed to bridge the gap between student engagement and administrative oversight. Unlike standard quiz platforms, QuizMaster implements a hierarchical learning structure where users must demonstrate proficiency in lower difficulty tiers (Easy, Moderate, Hard) before unlocking the **Advanced** level.

Built using the **MERN stack** (MySQL, Express.js, React, and Node.js), the system ensures a seamless, responsive user interface backed by a robust relational database. A key innovation of this project is the integration of an **Automated Certification Engine**. Upon successful completion of the highest difficulty tier, the system dynamically generates a downloadable PDF certificate, providing users with a formal validation of their domain mastery. This combination of gamified progression and tangible rewards creates a highly motivating ecosystem for modern learners.

3.1 DEFINE THE "PROBLEM"

Despite the proliferation of online educational platforms, several persistent challenges remain unaddressed in the development of lightweight, effective assessment systems. The "Problem" identified for this project can be categorized into the following technical and functional gaps:

- **Absence of Administrative Control:** Most open-access quiz applications focus solely on the end-user experience, leaving instructors or administrators without the tools to monitor progress or manage student records effectively.
- **Data Integrity and Clutter:** Without specialized management features like score resetting or user deletion, databases often accumulate redundant or obsolete data, which distorts performance analytics over time.
- **Engagement and Motivation Gaps:** Traditional digital quizzes often feel static and uninspiring; the lack of real-time visual feedback, such as countdown timers or progress indicators, reduces the sense of urgency and engagement required for effective learning.
- **Accessibility and Personalization:** A significant problem in current software is the "one-size-fits-all" UI. The lack of features like theme toggling (Dark/Light mode) can lead to eye strain and reduced accessibility for users during extended study sessions.

3.2 OBJECTIVES

To solve the problems, the **QuizMaster** project was developed with the following specific objectives:

- **Development of an Interactive Frontend:** To create a dynamic user interface using React.js that incorporates "Frosted Glass" design elements, responsive navigation, and real-time state management for quiz progression.
- **Implementation of Secure Backend Services:** To build a RESTful API using Node.js and Express that manages user authentication, secure data transmission, and administrative endpoint protection.
- **Establishment of a Relational Data Structure:** To design and implement a MySQL database schema featuring complex relationships between users, sub_categories, and user_progress tables for accurate tracking.
- **Creation of an Admin Analytics Suite:** To provide administrators with a high-level overview of system health through stats cards (Total Users, Average Scores) and a detailed student performance table.
- **Enhanced Data Management Features:** To empower admins with the ability to "Reset" specific topic scores or "Delete" entire user profiles, ensuring the database remains clean and relevant.
- **Automated Certification:** To integrate client-side PDF generation using the jsPDF library, allowing for the creation of dynamic certificates that pull user data directly from the system upon successful completion of the advanced level."

4. REVIEW OF LITERATURE

1. Introduction to the Domain

With the rapid growth of digital learning platforms, online quiz systems have become an important tool for knowledge assessment, self-evaluation, and skill development. Educational institutions and training organizations increasingly use web-based quiz applications to provide instant feedback, track learner performance, and support remote learning. Modern quiz systems are expected to support multiple subjects, difficulty levels, user authentication, and performance analytics to enhance the learning experience.

2. Study of Existing Systems

a) Traditional Online Quiz Systems

Several early web-based quiz platforms provide basic functionality such as:

Multiple-choice questions
Fixed question sets
Score calculation after submission

Limitation:
These systems usually lack domain categorization, adaptive difficulty levels, and detailed performance tracking.

b) Learning Management System (LMS) Based Quizzes

Popular LMS platforms like Moodle and Google Classroom include quiz modules with:

- Question banks
- Timed assessments
- Automatic grading

Limitation:
They are often complex, institution-dependent, and not focused on lightweight domain-wise practice or multi-level progression for individual learners.

c) Competitive Quiz and Practice Platforms

Modern platforms such as coding practice sites and exam preparation portals provide:

- Category-based quizzes
- Difficulty levels (easy, medium, hard)
- Limitation: Many of these systems are domain-specific (e.g., coding only) and do not provide customizable multi-domain quiz management.

3. Limitations of Existing Solutions

From the study of current systems, the following gaps are identified:

- Lack of combined domain-wise and multi-level structure in a single lightweight application.
- Limited role-based access (admin, user, evaluator).
- Insufficient performance analytics and progress tracking.
- Poor question management interface for administrators.
- Many systems are not scalable or customizable for different subjects.

4. Identified Research Gap

Existing quiz applications focus on:

- Simple online testing without difficulty progression, or
- Large LMS platforms that are complex and not user-friendly for small-scale learning.

There is a need for a domain-wise multi-level quiz system that:

- Organizes questions by subject/domain and difficulty level
- Provides secure login and role-based control
- Tracks user performance and progress
- Allows easy question insertion and management
- Offers a simple, responsive web interface

5. Need for the Proposed System

To overcome the above limitations, the proposed Domain-Wise Multi-Level Quiz Application is designed to:

- Support multiple domains such as HTML, CSS, JavaScript, etc.
- Provide four difficulty levels (easy, moderate, hard, advanced).
- Enable admin-side question management.
- Deliver instant scoring and result analysis.
- Improve self-learning and assessment efficiency for students.

This system aims to provide a simple, scalable, and user-friendly assessment platform suitable for academic mini-projects and practical learning environments.

5. STUDY AREA

5.1 LOCATION AND EXTENT

The development and primary deployment of the **QuizMaster** application are centered within the academic framework of **Pillai College of Arts, Commerce & Science (Autonomous)**, located in **New Panvel**. The extent of this study area encompasses the Department of Computer Applications, specifically targeting the student body and faculty involved in the academic year 2024-25.

The digital "extent" of the project reaches any local or networked environment where the Node.js server is hosted, providing access to students across the campus laboratory facilities. The study is bound by the specific quiz categories implemented in the system, which are designed to cater to the curriculum requirements of computer science students and general aptitude learners within the institution.

5.2 GENERAL DESCRIPTION OF STUDY AREA

The study area is characterized by a high-density academic environment where frequent assessment and rapid feedback are essential for student success. Pillai College, being a Re-accredited "A" Grade institution by NAAC, maintains a technologically advanced infrastructure that supports the implementation of modern web applications.

Within this study area, there is a distinct need for:

- **Digital Literacy Integration:** A community of students who are proficient in using web-based interfaces but require more organized platforms for self-evaluation.
- **Administrative Efficiency:** A department that handles large volumes of student performance data and requires automated tools to reduce manual record-keeping.
- **Controlled Evaluation Environments:** Laboratory settings where the "QuizMaster" application can be used to conduct mock tests or internal assessments under supervised conditions.
- **Scalable Learning Tools:** A study environment that is constantly evolving, necessitating software that can be updated with new topics and sub-categories without restructuring the entire database.

6. MATERIALS AND METHODS

6.1 MATERIALS USED

The materials used in the development of the QuizMaster project consist of the hardware infrastructure required to support full-stack web development and local server hosting.

- **Processor:** Intel Core i5 or higher (to handle concurrent execution of Node.js and React development servers).
- **RAM:** Minimum 8GB (required for smooth multitasking between VS Code, MySQL Workbench, and browser testing).
- **Storage:** 256GB SSD (for fast read/write operations during database transactions).
- **Input Devices:** Standard QWERTY Keyboard and Optical Mouse for coding and UI testing.

6.2 SOFTWARE USED

The software stack utilized for this project follows the MERN (MySQL, Express, React, Node) architecture, providing a scalable and modern development environment.

- **Operating System:** Windows 10/11 or macOS.
- **Frontend:** React.js for building the component-based User Interface.
- **PDF Generation:** jsPDF library for creating client-side certificates.
- **Backend:** Node.js and Express.js for RESTful API development.
- **Database:** MySQL for structured data storage and management.
- **Development Tool:** Visual Studio Code (VS Code) as the primary IDE.
- **Database Management:** MySQL Workbench or XAMPP for executing SQL queries.
- **API Testing:** Postman for verifying backend routes before frontend integration.

6.3 DATA PROCESSING

Data processing in QuizMaster involves the systematic collection of user responses and their conversion into meaningful performance analytics.

- **Data Input:** Users provide input through quiz selections and question responses.
- **Logic Processing:** The Node.js server calculates scores by comparing user answers against the correct options stored in the database.
- **Data Storage:** Final scores and level unlocks are updated in the user_progress table.

6.3.1 DIGITIZATION

Digitization refers to the process of converting traditional paper-based quiz questions and student records into a digital format.

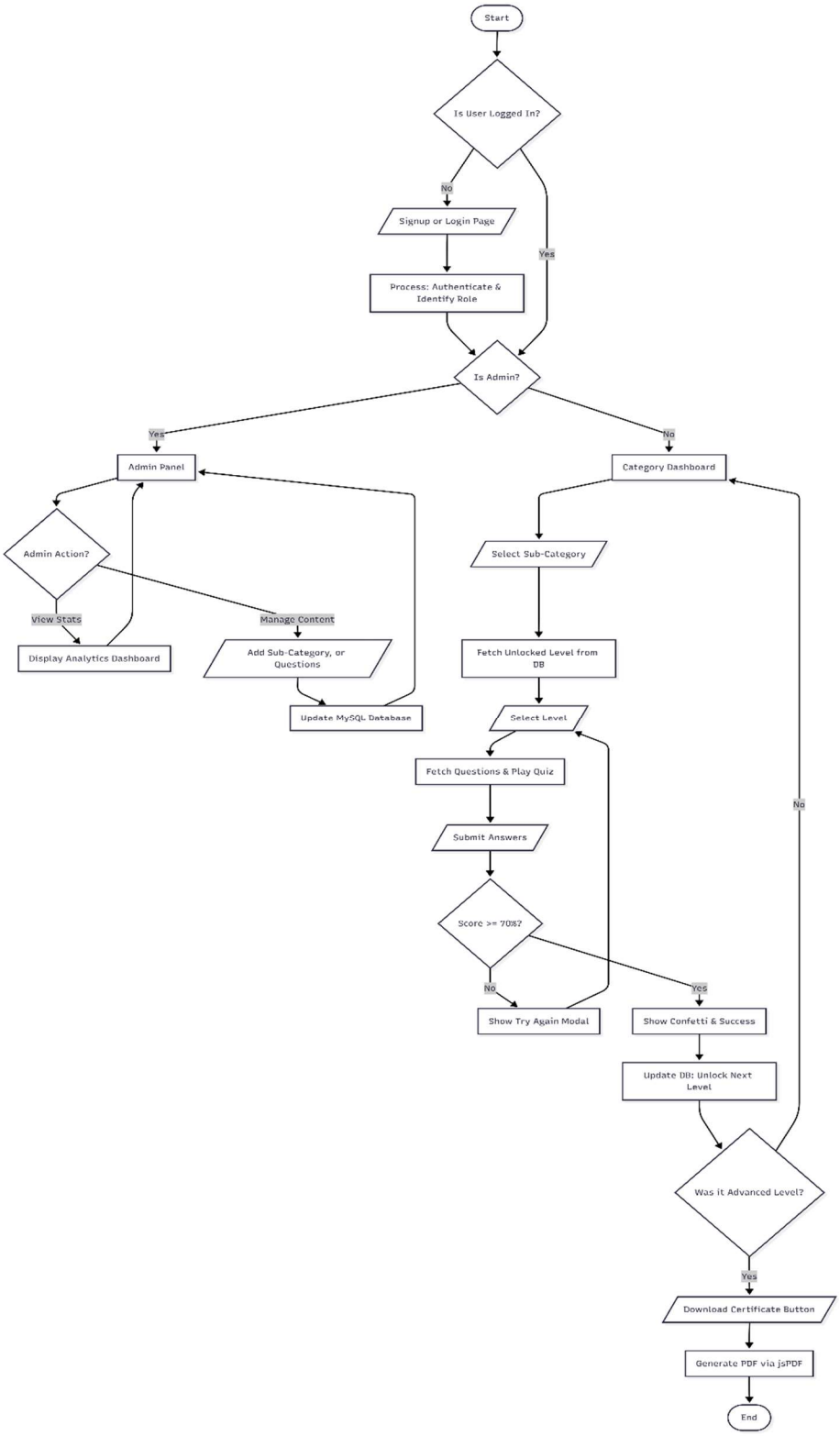
- **Content Conversion:** Physical question banks were digitized into JSON objects and SQL insert scripts.
- **Automated Record Keeping:** Manual grading is replaced by digital score calculation, ensuring 100% accuracy and immediate data availability.

6.4 RESEARCH METHODOLOGY

The research follows an **Agile Development Methodology**. This involves iterative cycles of coding, testing, and refining. The focus was on identifying the "Problem" of administrative oversight and developing a functional Dashboard to solve it. The project prioritized modularity, ensuring the "Theme Toggle" and "Admin Stats" could be developed and tested independently.

6.5 STEPS IN ANALYSIS (FLOWCHART)

The analysis of the system follows a logical sequence from user authentication to data visualization.

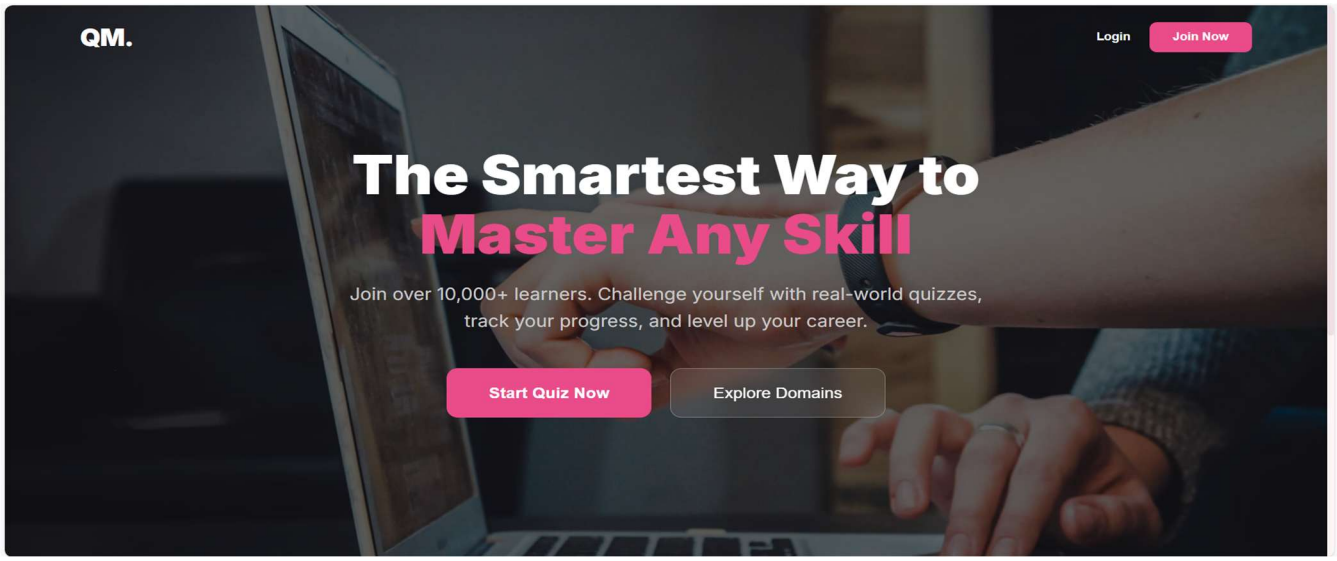


7. RESULTS

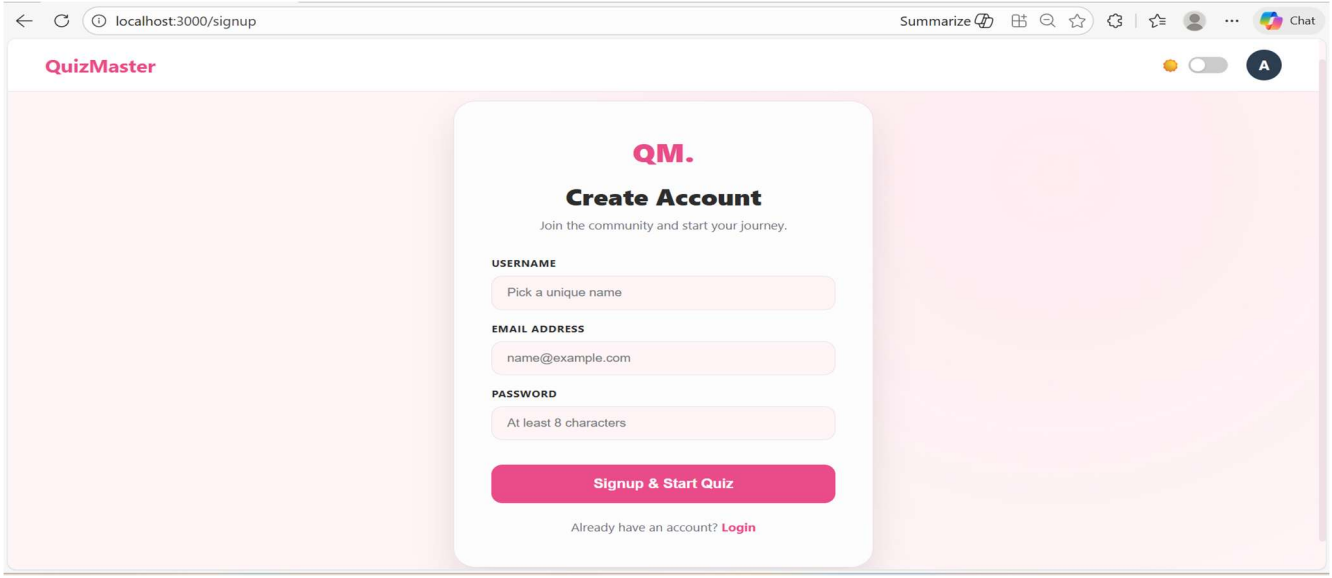
7.1 SYSTEM INTERFACE RESULTS

The primary result of this project is a fully functional, responsive web application. The interface successfully implements the "Frosted Glass" design language and supports dynamic theme switching.

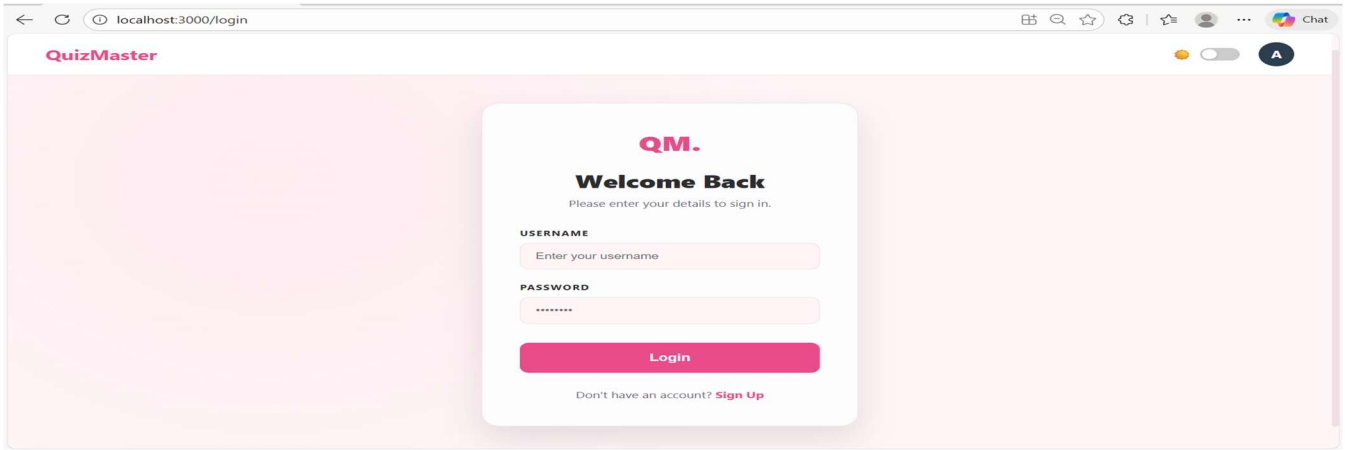
1. Landing page



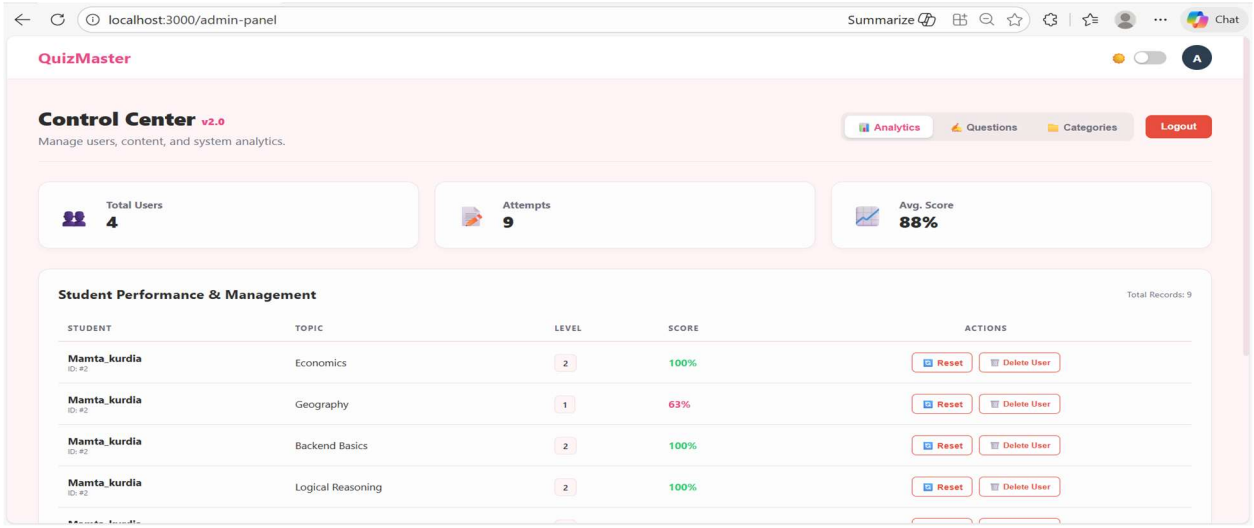
2. Signup page



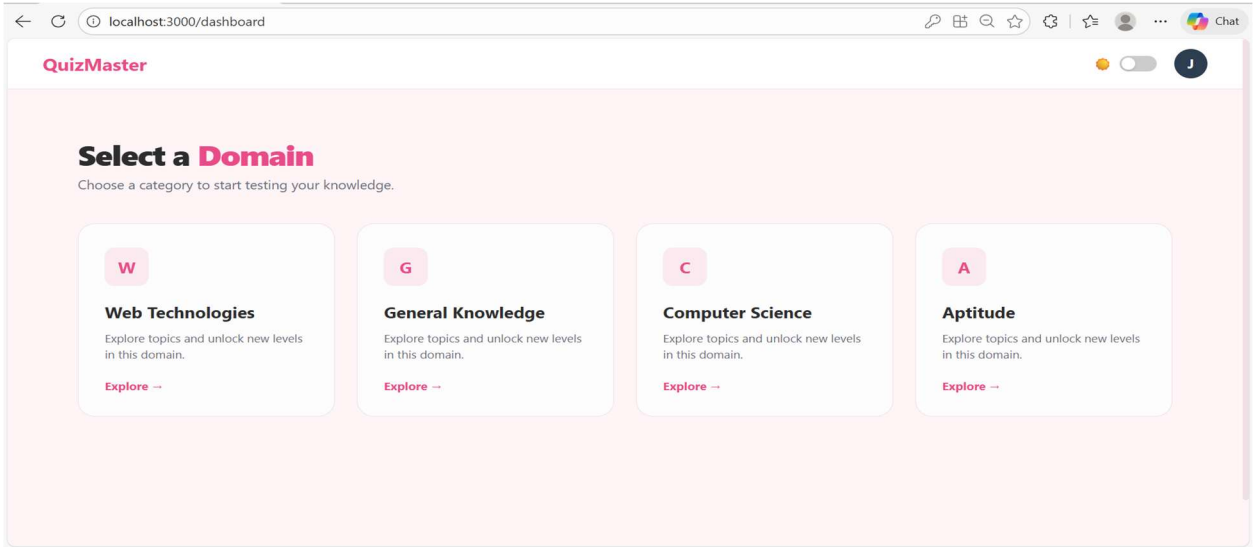
3. Login page



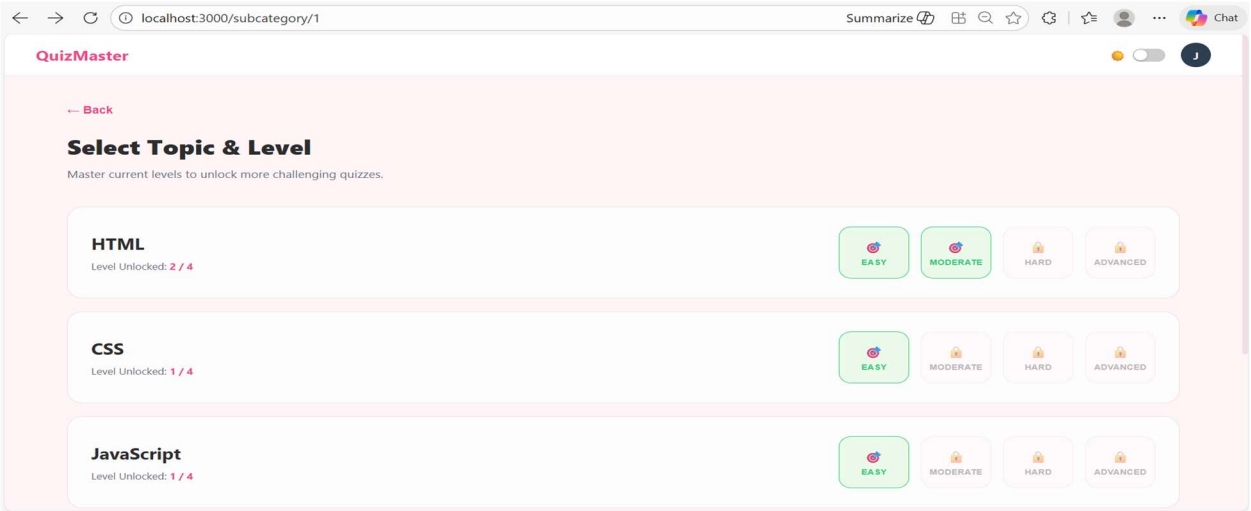
4. Admin-panel



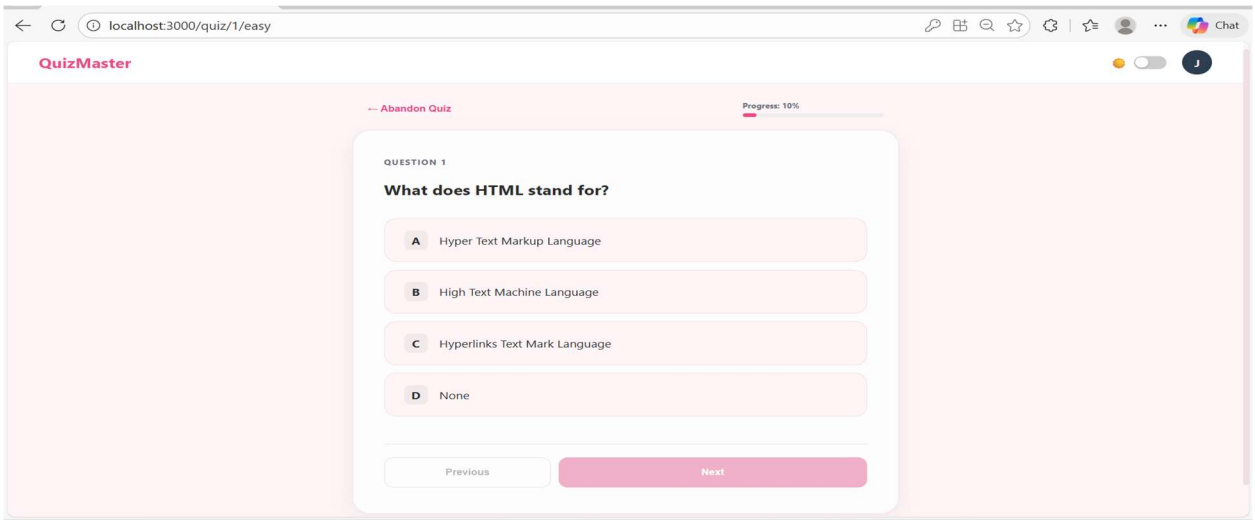
5. User Home page

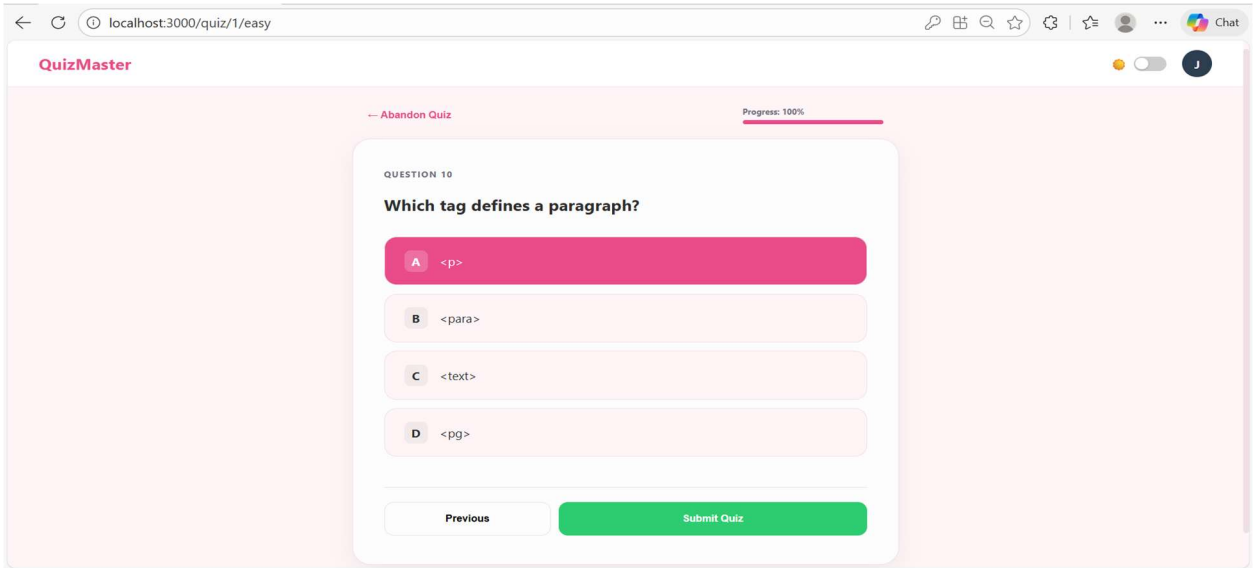


Sub-Category Page:

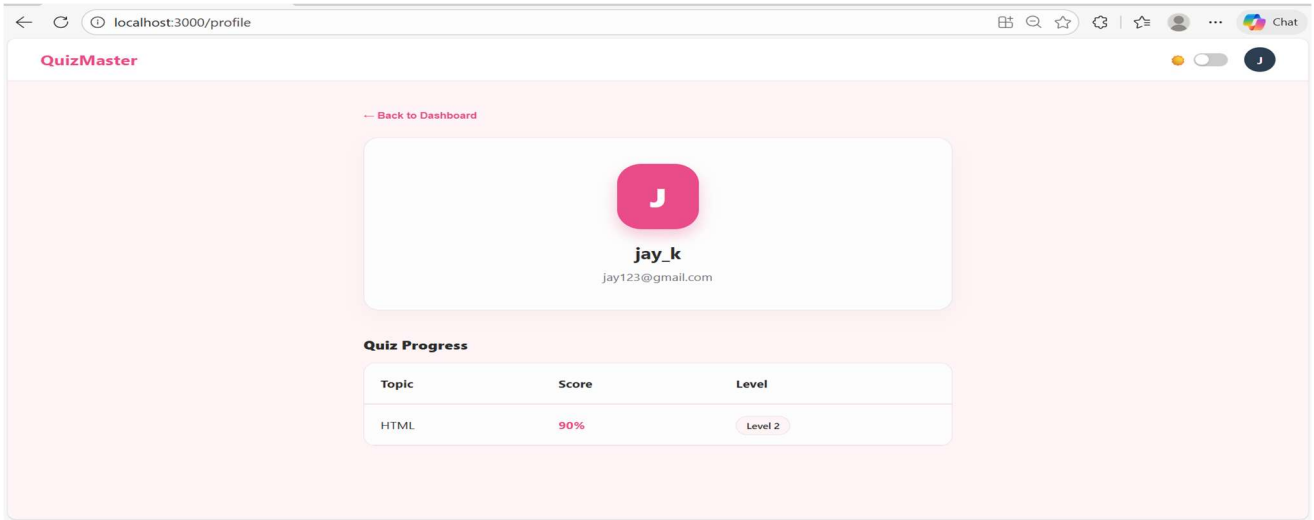


Quiz Page:

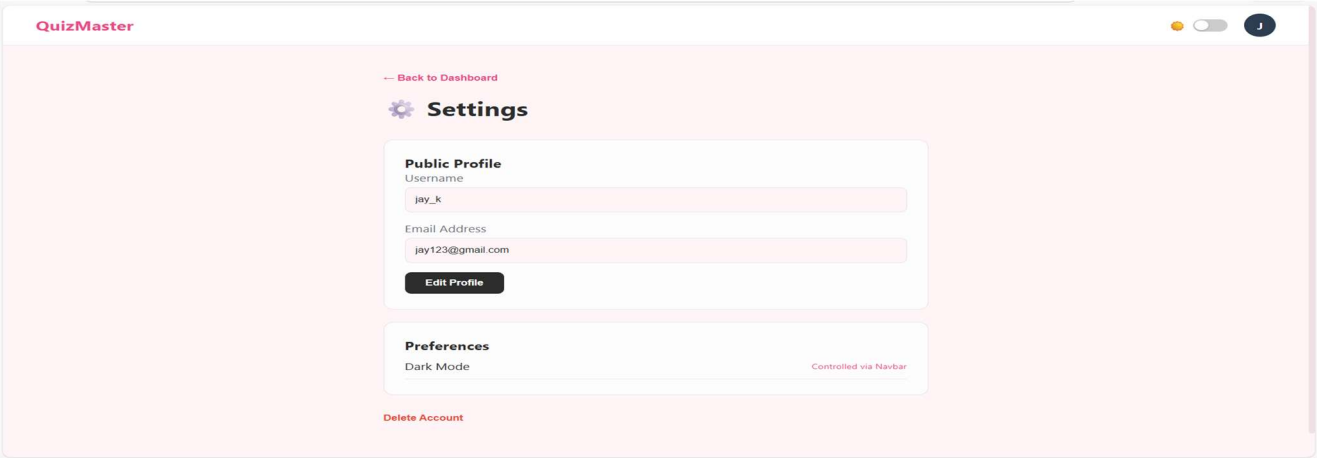




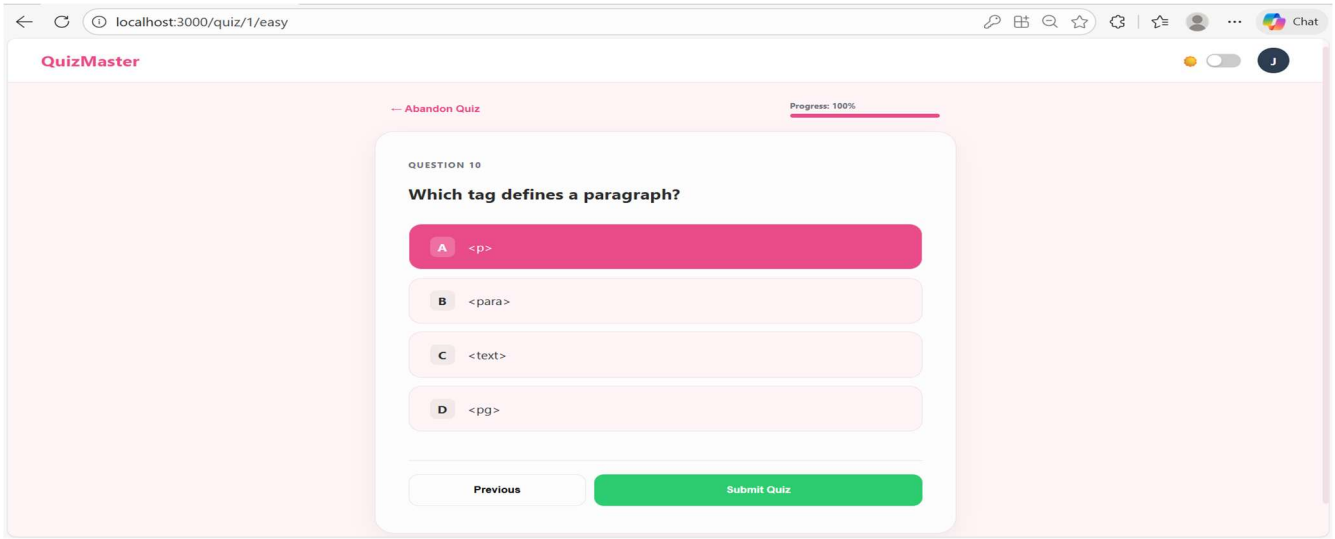
User Progress:



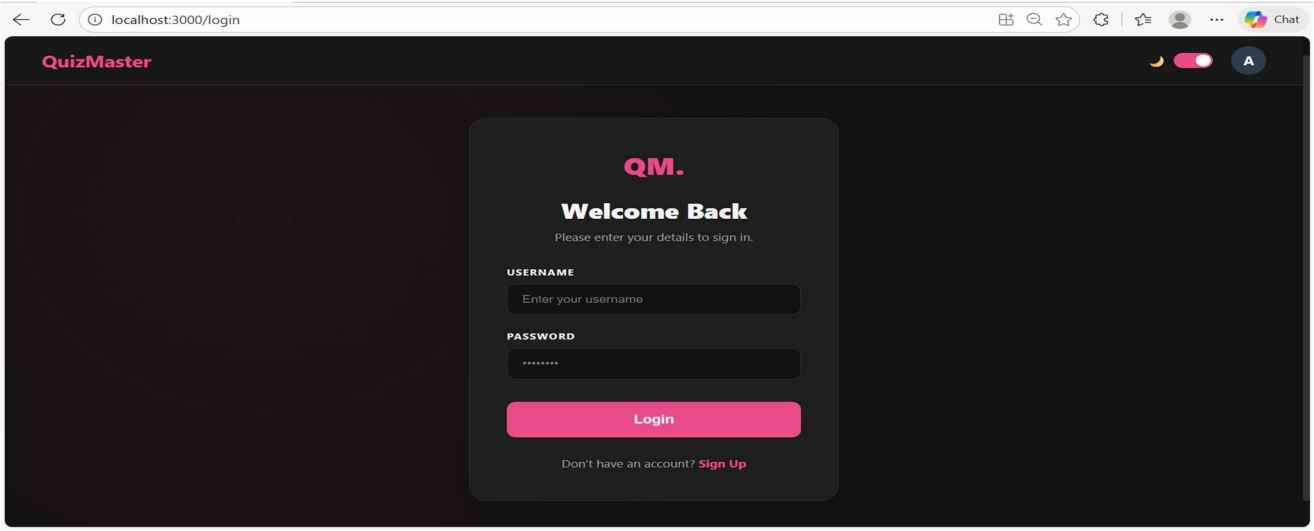
User Settings:



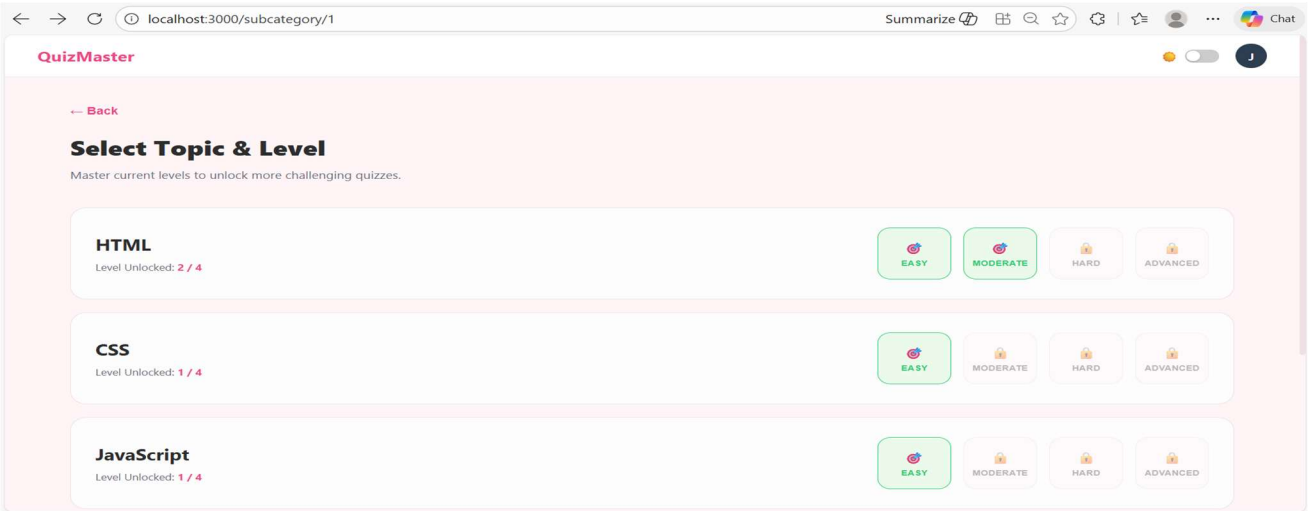
Can Review answers after submission:



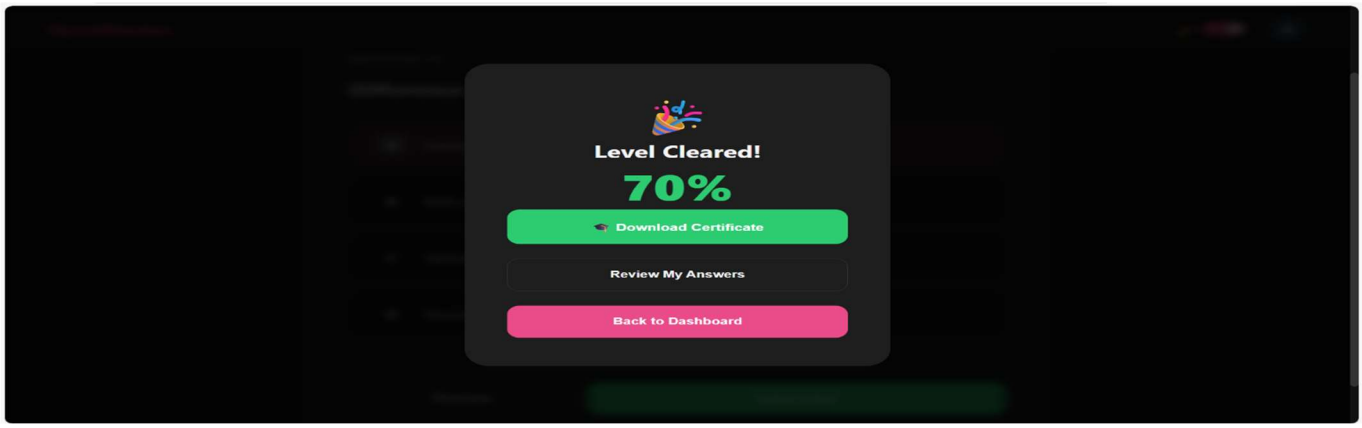
Theme Management: The implementation of a Dark/Light mode toggle ensures accessibility across different lighting conditions, with the system remembering the user's preference across sessions.

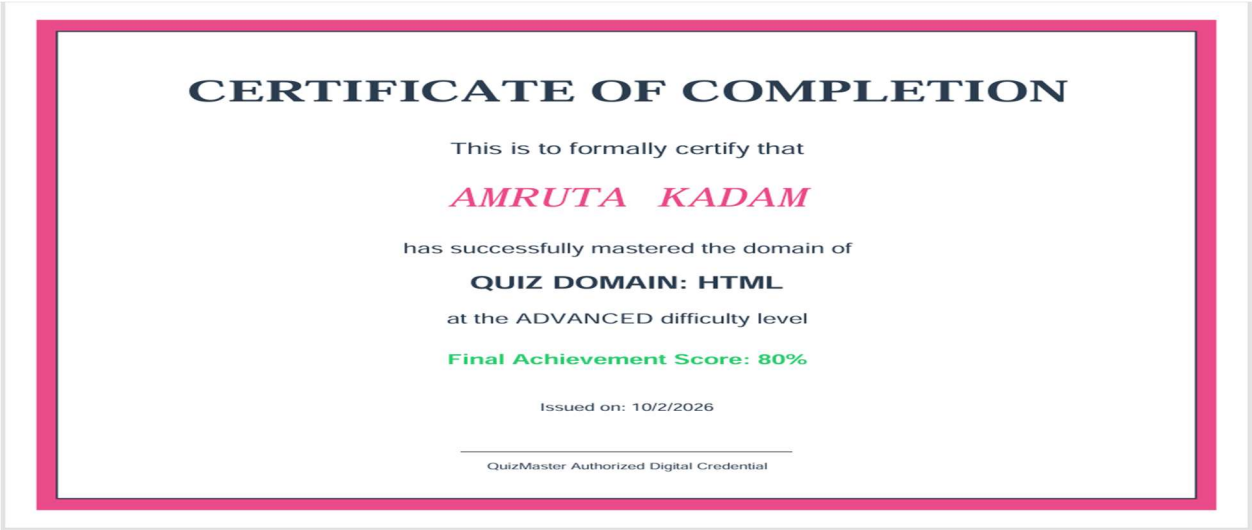


After submitting next level unlocked:



Certification Module: Upon mastering the Advanced level, the system triggers a 'Level Cleared' modal. For this specific tier, a 'Download Certificate' button is rendered, generating a high-quality PDF containing the student's name, the domain mastered, and the final score

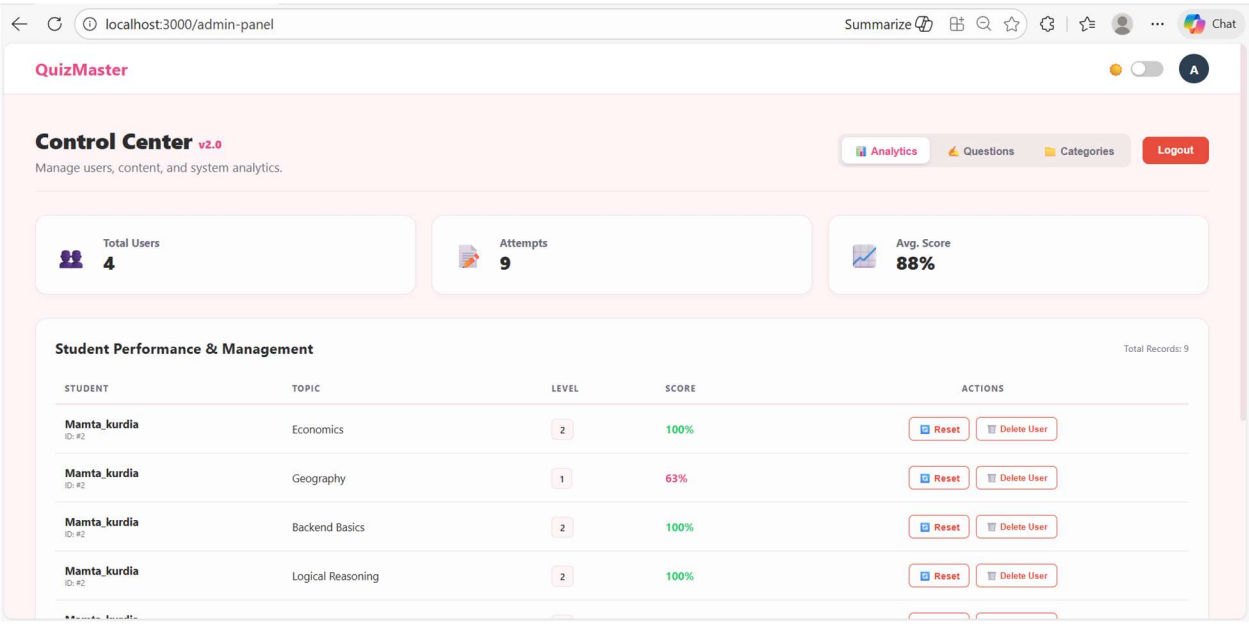




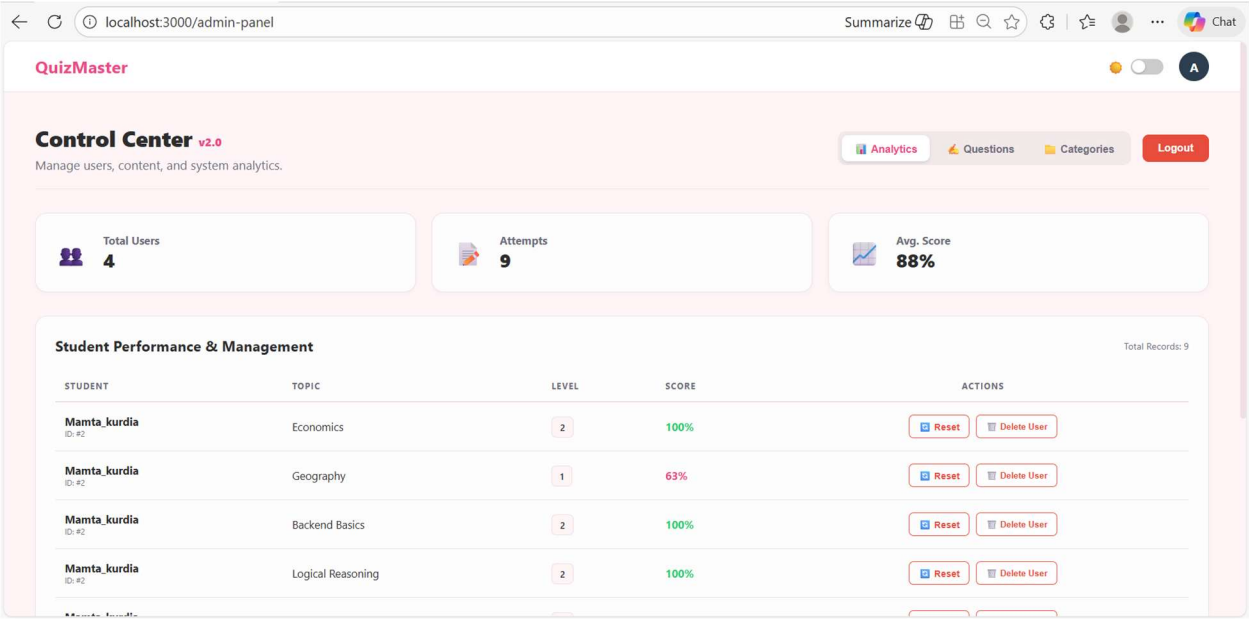
7.2 ADMINISTRATIVE DASHBOARD RESULTS

The Admin Panel serves as the central hub for data oversight. As shown in the "Student Performance" section, the system aggregates data from all users.

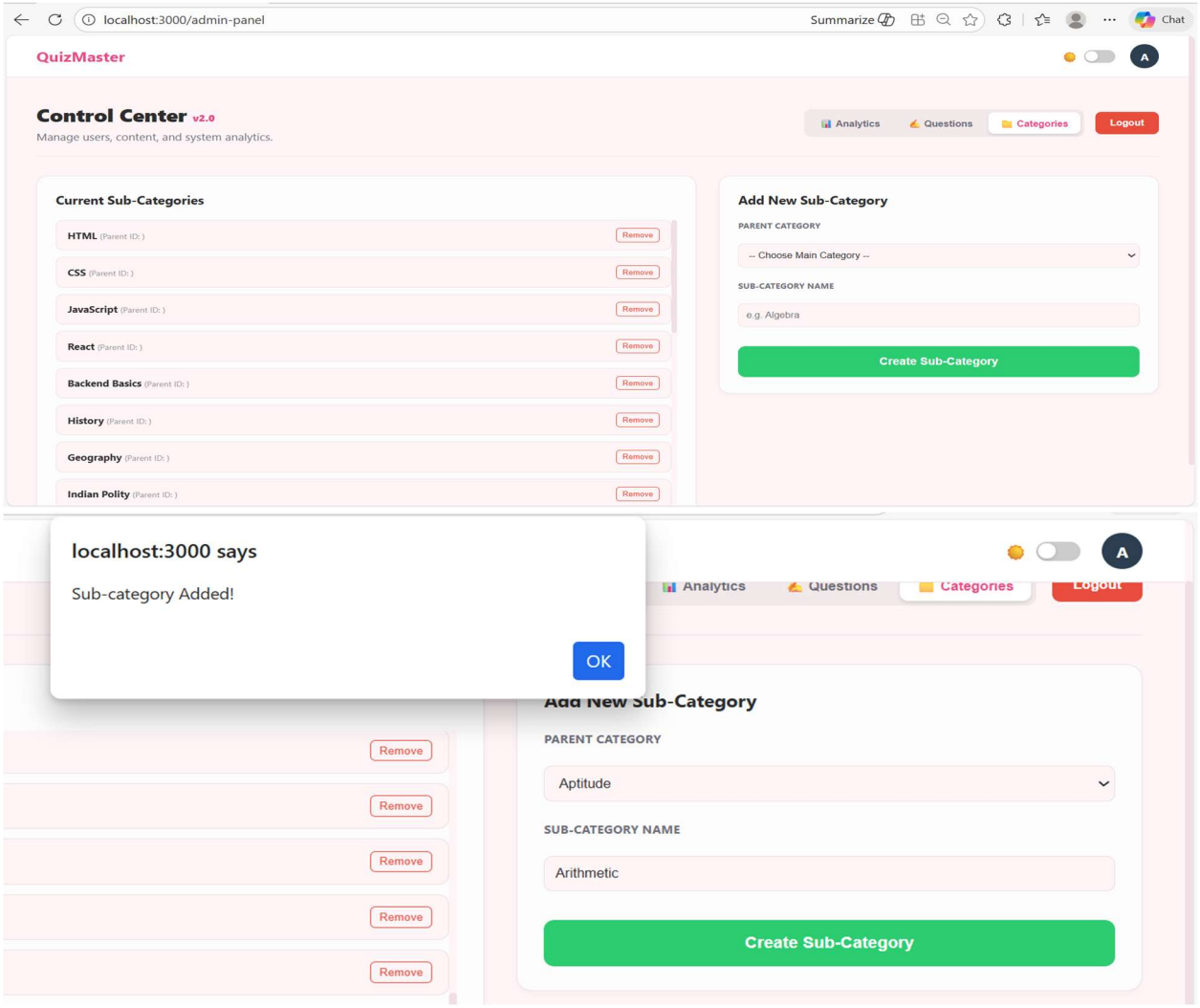
Real-time Analytics: The dashboard displays "Total Users," "Total Attempts," and "Average Score" using dynamic stat cards.

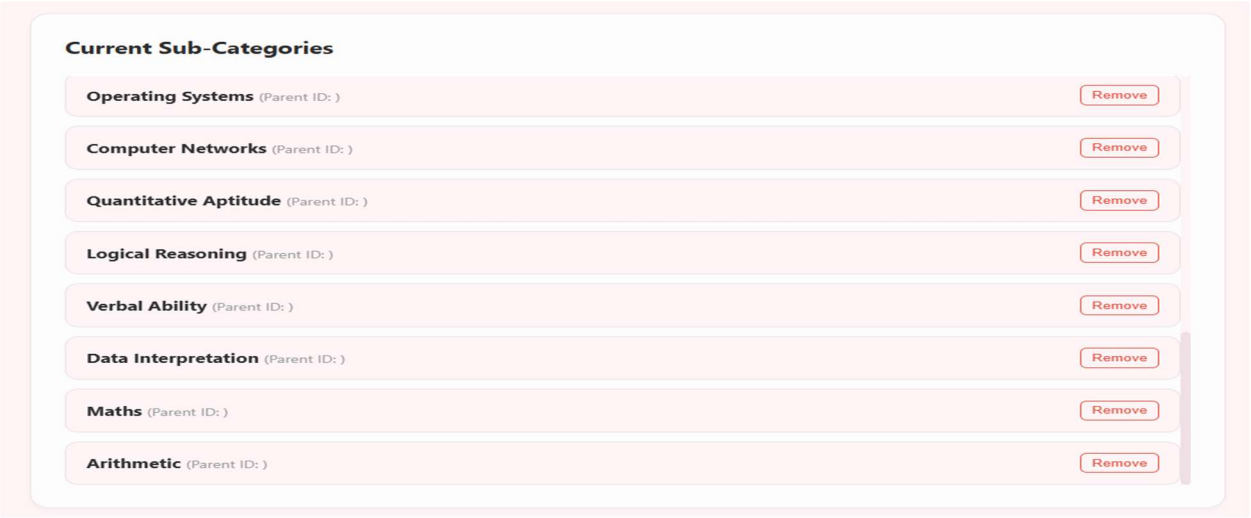


User Management Actions: The "Reset" and "Delete" functionalities were successfully tested. Deleting a user triggers a cascade effect, removing their progress data from the database to maintain storage efficiency.

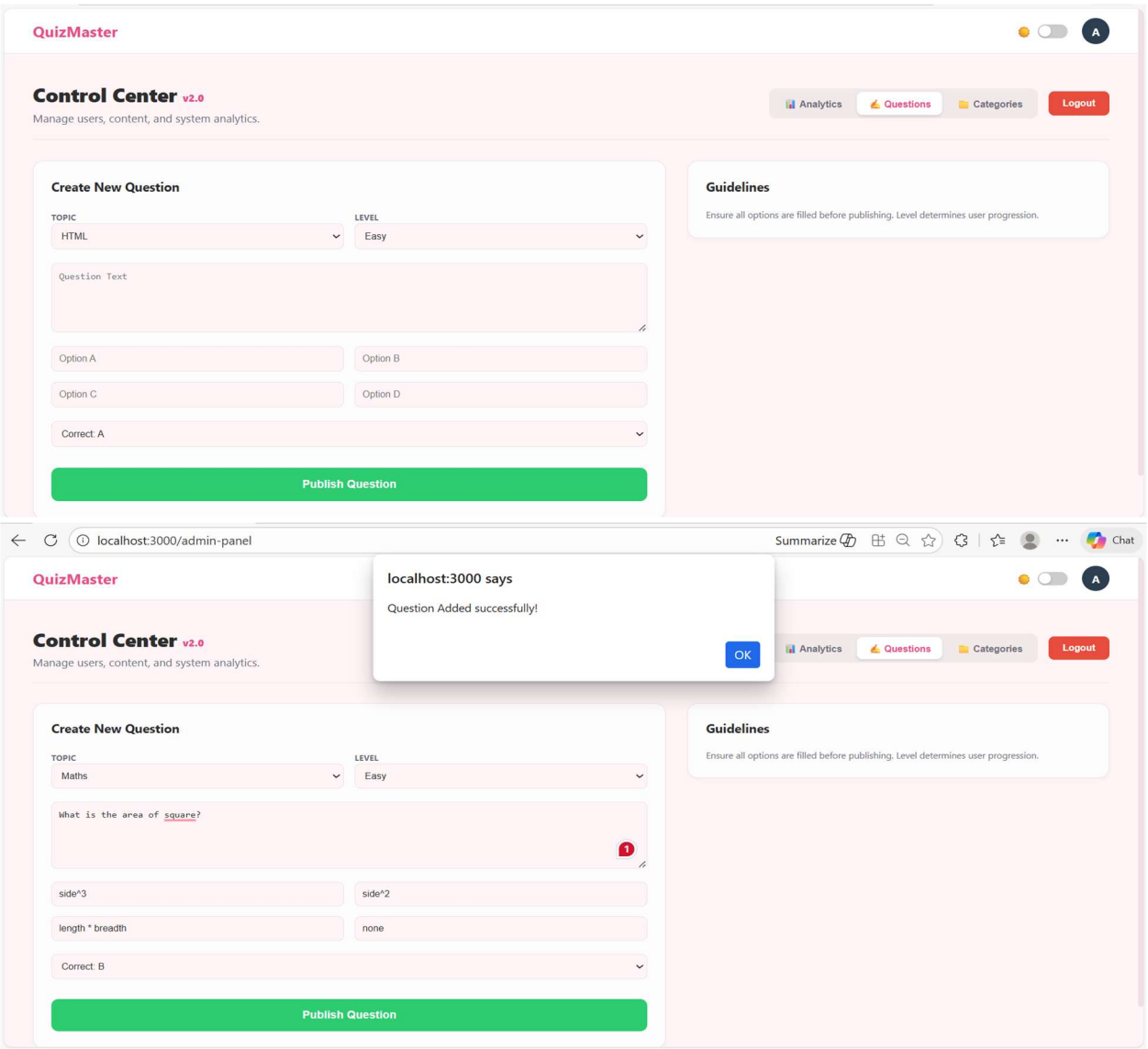


Adding new sub-category:





Adding new questions:



7.3 DATABASE RESULTS (DATA PERSISTENCE)

The MySQL database successfully maintains data integrity through relational mapping. Below are the resulting data states after system usage:

Table: Sample Output of user_progress Table

```
mysql> select * from user_progress;
+----+---------+-----------------+----------------+----------------+---------------------+
| id | user_id | sub_category_id | unlocked_level | highest_score | last_updated        |
+----+---------+-----------------+----------------+----------------+---------------------+
|  2 |       2 |               1 |              4 |             90 | 2026-02-04 13:07:10 |
|  6 |       2 |               2 |              2 |             70 | 2026-02-06 13:46:57 |
|  7 |       2 |              18 |              2 |            100 | 2026-02-06 14:14:46 |
|  8 |       2 |               6 |              2 |             75 | 2026-02-07 13:44:56 |
|  9 |       2 |              17 |              2 |            100 | 2026-02-07 13:55:27 |
| 10 |       2 |               5 |              2 |            100 | 2026-02-08 12:10:45 |
| 14 |       2 |               7 |              1 |             63 | 2026-02-08 14:04:59 |
| 17 |       2 |               9 |              2 |            100 | 2026-02-08 14:14:24 |
| 19 |      14 |               1 |              2 |             90 | 2026-02-08 20:43:13 |
+----+---------+-----------------+----------------+----------------+---------------------+
9 rows in set (0.03 sec)
```

Table: Sample Output of users Table

```
mysql> select * from users;
+----+-------------+---------------------------+--------------------------------------------------------------+-------+---------------------+
| id | username    | email                     | password_hash                                                | role  | created_at          |
+----+-------------+---------------------------+--------------------------------------------------------------+-------+---------------------+
|  1 | admin       | admin@quizapp.com         | $2b$10$cCy9qhMb77xebFOjbh7p..dN1wdGh8RDg7llpb.sjEiPnnIIJ41iG | admin | 2026-01-30 19:10:53 |
|  2 | Mamta_kurdia | mamta123@gmail.com        | $2b$10$oFO.Y460BPVVJieSzCu5reziB.RNCMW070Z8LMr6iGyl7IDetMAw. | user  | 2026-01-30 22:32:59 |
| 11 | sandhya_k   | sandhyakurdia2907@gmail.com | $2b$10$Eo44vMRRIEs4o5oVuCbmaOfBXxQNbPlCK76/VDjvuxdcQHWViQfUa | user  | 2026-02-08 19:45:44 |
| 12 | khushi_k    | khushi@gmail.com          | $2b$10$NcdX8bSamOkKffDfi4xg.u3ocHxJhW16amOxVSCrKlQs1/1PYlO8C | user  | 2026-02-08 19:59:40 |
| 14 | jay_k       | jay123@gmail.com          | $2b$10$afcN/16vKWM0RH8q9I6PGOq9YRnqj85foE/CGJlF6Uf2/mdThofRO | user  | 2026-02-08 20:05:11 |
+----+-------------+---------------------------+--------------------------------------------------------------+-------+---------------------+
5 rows in set (0.01 sec)
```

7.4 TEST CASE RESULTS

To verify the results, several test cases were conducted:

1. **Test Case 1 (Score Update):** A user completing a quiz with 80% correctly updated the highest_score column in MySQL.
2. **Test Case 2 (Unauthorized Access):** Attempting to access /admin without admin credentials successfully redirected the user to the login page.
3. **Test Case 3 (Deletion):** Clicking "Delete User" successfully removed the record from both the users and user_progress tables.

8. CONCLUSION

8.1 OVERALL CONCLUSION

The development of the **QuizMaster** application has successfully addressed the primary challenges identified in traditional digital assessment tools. By utilizing the MERN stack (MySQL, Express, React, Node.js), the project has delivered a high-performance, responsive platform that caters to both students and administrators.

The student-facing interface provides a gamified experience with real-time feedback, timers, and theme customization, which significantly enhances user engagement and learning retention. Simultaneously, the Admin Dashboard provides a robust solution for data oversight, allowing for efficient monitoring of student performance through analytics and management tools like score resetting and user account deletion. The integration of a relational database (MySQL) ensures that all data is stored securely and remains consistent across various user sessions. In conclusion, QuizMaster serves as an effective, scalable, and modern educational tool that simplifies the evaluation process while providing a personalized experience for the learner. The inclusion of a certification module adds a professional dimension to the platform, rewarding student persistence and providing a tangible proof of domain mastery.

8.2 FUTURE DEVELOPMENTS

While the current version of QuizMaster provides a solid foundation for quiz management, several enhancements can be integrated in future iterations to further increase its utility:

- **Advanced Analytics:** Implementing data visualization charts (using libraries like Chart.js) to show student performance trends over weeks or months.
- **Leaderboard and Social Integration:** Adding a global leaderboard to foster healthy competition among students and allowing results to be shared on professional platforms like LinkedIn.
- **Multi-Media Quizzes:** Expanding the question database to include images, audio clips, and video-based questions for a more comprehensive assessment.
- **AI-Generated Questions:** Integrating OpenAI or similar APIs to automatically generate unique quiz questions based on a specific topic or uploaded PDF textbook.
- **Automated Email Reports:** Developing a feature that automatically emails performance summaries to both students and faculty advisors upon completion of a level.
- **Mobile Application:** Converting the current web-based platform into a cross-platform mobile app (using React Native) for learning on the go.

9. REFERENCES AND BIBLIOGRAPHY

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2. WEB RESOURCES

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4. https://expressjs.com/en/api.html (Express.js Framework API Reference)
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